End of Chapter Problems -

Chapter 1, Sections 1.1 - 1.3

Probability Book, Bertsekas and Tsitsiklis

Problem 1.

$\Omega = \{1,2,3,4,5,6\}$; $A = \{2,4,6\}$, $B = \{4,5,6\}$

$A \cup B = \{2,4,5,6\} \Rightarrow (A \cup B)^c = \{1,3\}$

$A^c = \{1,3,5\}$, $B^c = \{1,2,3\} \Rightarrow A^c \cap B^c = \{1,3\}$

So, $(A \cup B)^c = A^c \cap B^c$

$A \cap B = \{4,6\} \Rightarrow (A \cap B)^c = \{1,2,3,5\}$

$A^c \cup B^c = \{1,2,3,5\}$

So, $(A \cap B)^c = A^c \cup B^c$

Problem 2.

(a) $A^c = A^c \cap \Omega = A^c \cap (B \cup B^c) = (A^c \cap B) \cup (A^c \cap B^c)$

In the same way, $B^c = (B^c \cap A) \cup (B^c \cap A^c)$

(b) $(A \cap B)^c = A^c \cup B^c$ (De Morgan's Laws)

$A^c = (A^c \cap B) \cup (A^c \cap B^c)$

$B^c = (B^c \cap A) \cup (B^c \cap A^c)$ } proved in (a)

$A^c \cup B^c = (A^c \cap B) \cup (B^c \cap A) \cup (A^c \cap B^c)$ (above)

Therefore, $(A \cap B)^c = (A^c \cap B) \cup (B^c \cap A) \cup (A^c \cap B^c)$

(c) $\Omega = \{1, 2, 3, 4, 5, 6\}$

$A = \{1, 3, 5\}$, $B = \{1, 2, 3\}$

$A \cap B = \{1, 3\} \Rightarrow (A \cap B)^c = \{2, 4, 5, 6\}$

$A^c = \{2, 4, 6\}$, $B^c = \{4, 5, 6\}$

$A^c \cap B = \{2\}$, $B^c \cap A = \{5\}$, $A^c \cap B^c = \{4, 6\}$

$\Rightarrow (A^c \cap B) \cup (B^c \cap A) \cup (A^c \cap B^c) = \{2, 4, 5, 6\}$

$\Rightarrow (A \cap B)^c = (A^c \cap B) \cup (B^c \cap A) \cup (A^c \cap B^c)$

Problems 3 and 4 are solved in the textbook. That is, solutions are given there.

Problem 5. A = selected student is genius

B = selected student loves chocolate

Given $P(A) = 0.6$, $P(B) = 0.7$, $P(A \cap B) = 0.4$

$A^c \cap B^c$ = selected student is neither genius nor loves chocolate.

$$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.9$$

$$\Rightarrow P(A^c \cap B^c) = 1 - 0.9 = 0.1$$

Problem 6. Let x_o = probability of any odd face

x_e = probability of any even face

$x_e = 2x_o$ and $3x_e + 3x_o = 1 \Rightarrow 9x_o = 1$

$\Rightarrow x_o = 1/9$, $x_e = 2/9$.

$$P(\{1,2,3\}) = P(\{1\}) + P(\{2\}) + P(\{3\})$$

$$= x_o + x_e + x_o = 4/9.$$

Problem 7. Experiment stops when we get a 2 or a 4.

$\Rightarrow$ Sample space $= \{1, 2, 3, \ldots\}$ = set of natural nos.

= # rolls till we get a 2 or a 4.

Problem 8. W_i = event that I win with ith opponent

L_i = event that I lose with ith opponent

All possible outcomes:

$W_1W_2W_3$, $W_1W_2L_3$, $W_1L_2W_3$, $W_1L_2L_3$, $L_1L_2L_3$, $L_1W_2W_3$, $L_1W_2L_3$, $L_1L_2W_3$

Favorable outcomes: $W_1W_2W_3$, $W_1W_2L_3$, $L_1W_2W_3$

since out of 3 games, I must win 2 in a row, I cannot have a L_2. Therefore, it is optimal to have the weakest player in the 2nd turn.

Problem 9. (a) S_i s form a partition of Ω.

$$\Rightarrow A = A \cap \Omega = A \cap (\cup_{i=1}^{n} S_i) = \cup_{i=1}^{n} (A \cap S_i)$$

since $S_i \cap S_j = \phi \ \forall i,j \Rightarrow (A \cap S_i) \cap (A \cap S_j) = \phi \ \forall i,j$

$$\Rightarrow P(A) = P(\cup_{i=1}^{n} (A \cap S_i)) = \sum_{i=1}^{n} P(A \cap S_i)$$

(b) $B^c \cap C$, $B \cap C^c$, $B \cap C$, and $B^c \cap C^c$ form a partition of Ω.

$$\Rightarrow P(A) = P(A \cap B^c \cap C) + P(A \cap B \cap C^c) + P(A \cap B \cap C) + P(A \cap B^c \cap C^c)$$

Now, $A \cap B = (A \cap B \cap C) \cup (A \cap B \cap C^c)$

$A \cap C = (A \cap B \cap C) \cup (A \cap B^c \cap C)$

$\Rightarrow P(A \cap B) = P(A \cap B \cap C) + P(A \cap B \cap C^c)$

$P(A \cap C) = P(A \cap B \cap C) + P(A \cap B^c \cap C)$

$\Rightarrow P(A) = P(A \cap C) - P(A \cap B \cap C) +$
$P(A \cap B) - P(A \cap B \cap C) + P(A \cap B \cap C) +$
$P(A \cap B^c \cap C^c)$

$\Rightarrow P(A) = P(A \cap B) + P(A \cap C) + P(A \cap B^c \cap C^c) -$
$P(A \cap B \cap C)$

Problem 10. $A = (A \cap B) \cup (A \cap B^c)$

$B = (A \cap B) \cup (A^c \cap B)$

$\Rightarrow P(A) = P(A \cap B) + P(A \cap B^c)$

$P(B) = P(A \cap B) + P(A^c \cap B)$

$\Rightarrow P((A \cap B^c) \cup (A^c \cap B)) = P(A \cap B^c) + P(A^c \cap B)$
$= P(A) + P(B) - 2P(A \cap B)$

Problems 11-13 are solved in the textbook

Problem 14. (a) Probability of doubles $= 6/36 = 1/6$

(b) Sum $= \{4, 3, 2, 1\}$

Doubles can be rolled if sum $= 4$ or 2

Sum $= 4 \rightarrow (2,2), (1,3), (3,1)$

Sum $= 2 \rightarrow (1,1)$

Favorable outcomes $= 4$

Total outcomes $= 6$

Probability of doubles given sum $\leq 4 = 4/6 = 2/3$

(c) Favorable cases: $6 + 6 - 1 = 11$

Probability of at least 1 six $= 11/36$

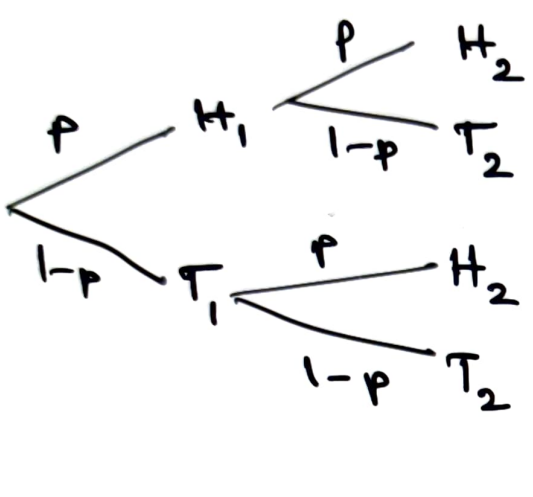
(d) # Total outcomes $= 30$ $\quad (36 - 6$ doubles$)$

Outcomes with exactly one 6 $= 10$

Probability that one die is a 6 (given no doubles)

$= 10/30 = 1/3.$

Problem 15. Let probability of head = p.



$$P(H_1H_2) = p^2$$

$$P(H_2 \mid H_1) = p$$

$$P(H_1H_2 \mid \{H_1H_2, H_1T_2, T_1H_2\}) = \frac{p^2}{p^2 + 2p(1-p)}$$

Now whether $P(H_2 \mid H_1) \geq P(H_1H_2 \mid \{H_1H_2, H_1T_2, T_1H_2\})$

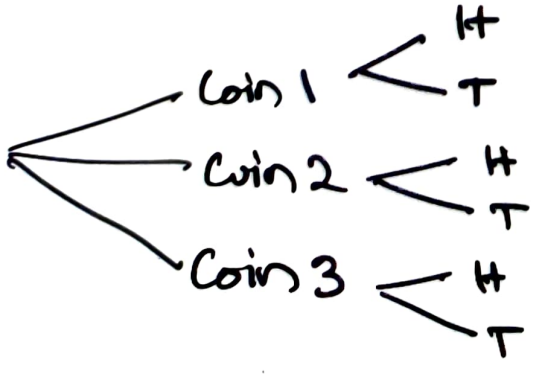
depends on p.

For a fair coin, $p = 0.5$ and

$P(H_2 \mid H_1) = 0.5$

$P(H_1H_2 \mid \{H_1H_2, H_1T_2, T_1H_2\}) = 0.33$

Problem 16.



$P(H \mid \text{Coin } 1) = 1$

$P(H \mid \text{Coin } 2) = 0.5$

$P(H \mid \text{Coin } 3) = 0$

$P(\text{Coin } 1) = P(\text{Coin } 2) = P(\text{Coin } 3) = 1/3$

$\Rightarrow P(H) = 0.5$

$$\Rightarrow P(\text{Coin 2} \mid H) = \frac{P(\text{Coin 2} \cap H)}{P(H)} = \frac{\frac{1}{3} \cdot 0.5}{0.5} = \frac{1}{3}$$

Problem 17.

ways to choose 4 items from 100 items = $^{100}C_4$

ways to choose 4 items from 95 items = $^{95}C_4$
(all perfect items)

$$\Rightarrow \text{Probability} = \frac{^{95}C_4}{^{100}C_4} = \frac{95.94.93.92}{100.99.98.97}$$

Sequence Approach : A_i = event that item i is fine

$$P(A_1 \cap A_2 \cap A_3 \cap A_4) = P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1, A_2) \cdot P(A_4 \mid A_1, A_2, A_3)$$

$$P(A_1) = \frac{95}{100}, \quad P(A_2 \mid A_1) = \frac{94}{99},$$

$$P(A_3 \mid A_1, A_2) = \frac{93}{98}, \quad P(A_4 \mid A_1, A_2, A_3) = \frac{92}{97}.$$

$$\Rightarrow P(A_1 \cap A_2 \cap A_3 \cap A_4) = \frac{95.94.93.92}{100.99.98.97}$$

Problem 18. $(A \cap B) \cap B = A \cap B$

$$\Rightarrow P(A \cap B \mid B) = \frac{P(A \cap B)}{P(B)} = P(A \mid B)$$